Practical

Intro to Flask: Building an Interactive Web App

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What is Flask?

Flask is a Python-based web framework that allow developers to create a web interface by connecting frontend user interfaces with backend Python logic.

Flask + python™ = **An interactive web application**

Without Flask

```
C:\Windows\System32\cmd.e

Microsoft Windows [Version 10.0.26200.8037]
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C:\Users\lisbe\OneDrive\Attachments\flask demo>python simplepython.py
Enter your name: Lis
Hello, Lis

C:\Users\lisbe\OneDrive\Attachments\flask demo>
```
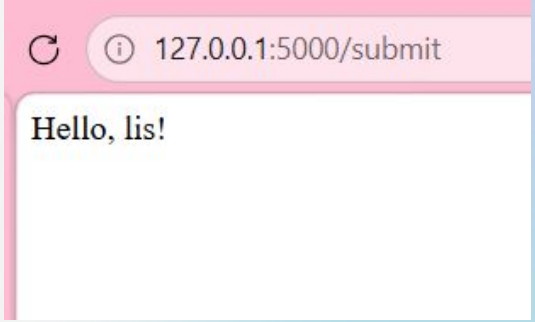

With Flask

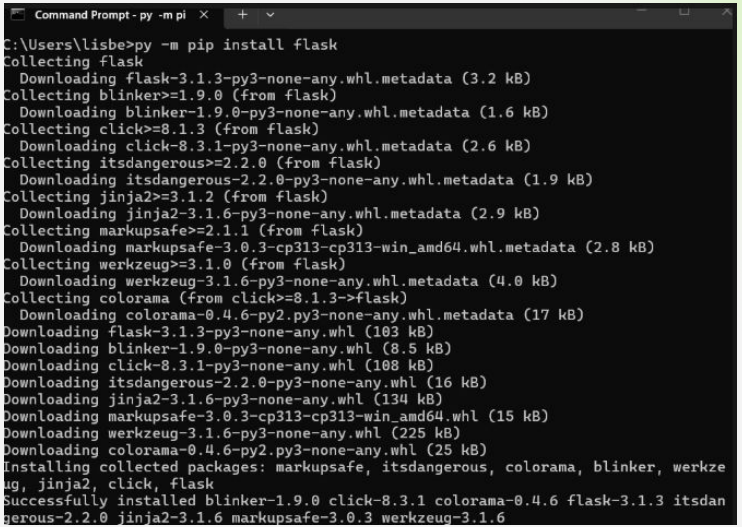


Why Flask?

- ★ Better User Experience
  - No terminal needed
  - User-friendly
- ★ Accessibility
  - Anyone can access it through a browser
  - Requires no coding knowledge
- ★ Scalability
  - Is able to handle more complex knowledge
    - Connects components: user input, CSVs, logic
  - Can connect to databases, APIS, etc
  - More suitable for real-world applications

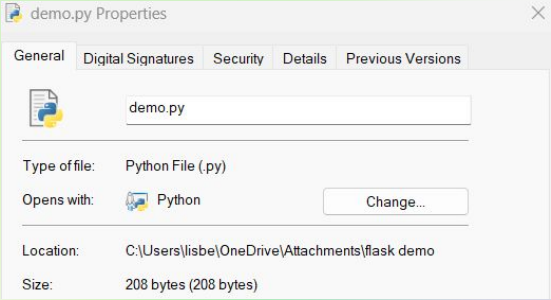
How to install Flask?

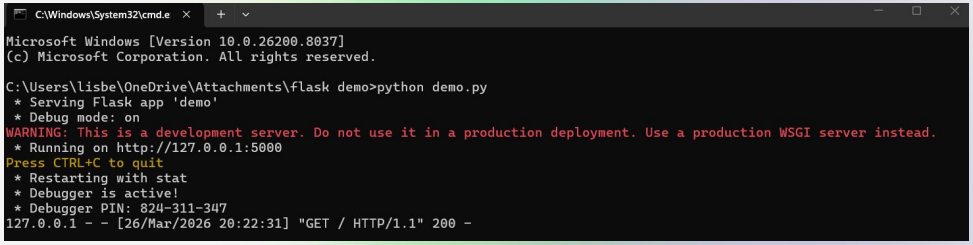
1. Install Python
   a. Download Python | Python.org
      b. Python 3.13.12
      c. “Add Python to Path”
2. Open Terminal
3. Install Flask
      a. “pip install flask”

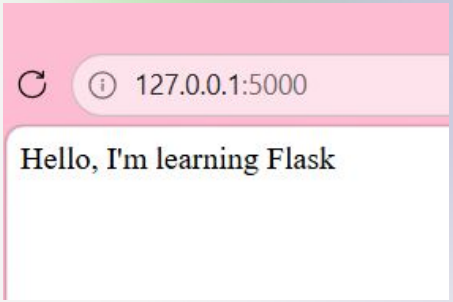


What is needed to build a Flask App w User Interface?

1. A Python File: This will run Flask
   a. Flask can run without the need of an HTML File, but if there's no HTML file there will be no way for it to collect user input

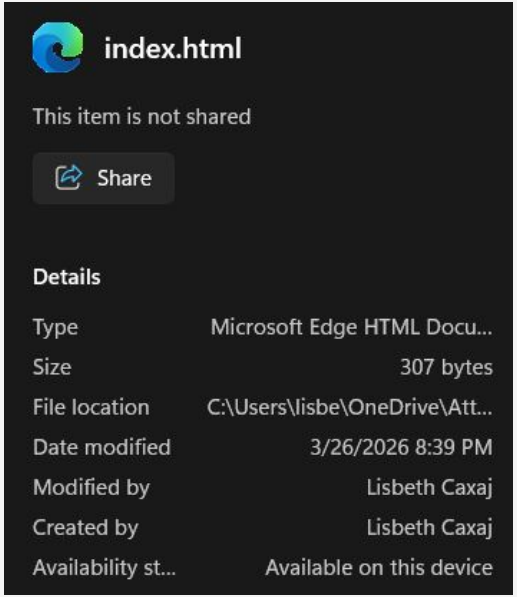


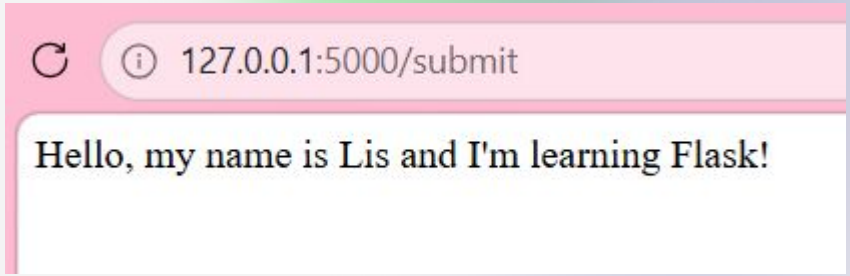




2. HTML File: this collects user input

- Used when we want to build interactive interfaces





File Structure

- ★ Main Project Folder (flask demo)
  - Python File (demo.py)
- ★ templates Folder (default)
  - HTML File (index.html)

Flask Demo	**Main Folder for Application**	flask demo
Python File	**Handles application logic**	demo.py
templates	**Stores HTML files**	templates
HTML File	**Webpage the user interacts with**	index.html

Demo.py

```
from flask import Flask, render_template, request
```

imports flask

loads HTML files

allows us to access user input

```
app = Flask(__name__)
```

creates our flask application

```
@app.route("/")
def home():
    return render_template("index.html")
```

defines home page & loads and displays the html file

```
@app.route("/submit", methods=["POST"])
def submit():
    name = request.form["name"]
    return f"Hello, my name is {name} and I'm learning Flask!"
```

route runs when the user submits the form

method signifies we're sending data from user to server

takes input user typed, stores in in a variable

Flask uses input & returns a dynamic message to user

```
if __name__ == "__main__":
    app.run(debug=True)
```

starts flask server

Index.html

File Edit View

```
<!DOCTYPE html>
<html>
<head>
    <title>Flask Demo</title>
</head>
<body>

    <h1>Enter Your Name</h1>

    <form action="/submit" method="POST">
        <input type="text" name="name" placeholder="Enter your name">
        <button type="submit">Submit</button>
    </form>

</body>
</html>
```

Basic HTML Structure

heading of the page

when user hits submit, data is sent to submit route using the post request

where user types their name: flask uses name attribute to retrieve the input

sends the form data to flask when clicked

closes HTML structure